

# Beta-Sorted Portfolios

## Supplemental Appendix: Applied

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### Abstract

This supplemental appendix provides practical guidance and numerical evidence for beta-sorted portfolio estimation and inference. Part [III](#) presents data-driven choices of the first-step bandwidth and the second-step number of portfolios and discusses their use in pointwise-in- $\beta$  and uniform-in- $\beta$  inference. Part [IV](#) reports Monte Carlo evidence on confidence interval coverage and average length, additional empirical results for the business-credit factor, and an application to capital share risk. The companion Supplemental Appendix: Theory provides the theoretical foundations for these estimation and inference procedures.

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## Part III: Tuning Parameter Selection

This part turns the rate restrictions in Part II into implementable tuning choices for the classical beta-sorted portfolio estimator. Section SA-III.1 gives the pointwise first-step plug-in rule for the bandwidth  $h$  and the integrated-MSE plug-in rule for the second-step number of portfolios  $J$ . Section SA-III.2 keeps those  $h$  and  $J$  choices, and proposes a data-driven selector for the near-origin truncation parameter  $\delta$ . Section SA-III.3 records the separate implication for factor-cancelling linear contrasts, whose idiosyncratic normalization requires dedicated undersmoothing rather than the default plug-in choices.

### SA-III.1 Pointwise-in- $\beta$

This section summarizes a practical data-driven tuning rule for the classical beta-sorted portfolio estimator and an associated robust bias-corrected confidence interval. Throughout this section, braces on tuning parameters denote the relevant collection over time and units; for example,  $\{J_t\}$  denotes  $\{J_t : t = H + 1, \dots, T\}$ .

- We write the first-step estimator as  $\widehat{\beta}_{it}(\mathbf{p}, h_{it})$ , where  $\mathbf{p}$  is the local-polynomial order and  $h_{it}$  is the bandwidth.
- We write the period- $t$  second-step fit as  $\widehat{M}_t(\beta; \mathbf{q}, J_t)$ , where  $\mathbf{q}$  is the portfolio polynomial order and  $J_t$  is the number of portfolios per beta dimension at time  $t$ .

The full second-step estimator averages these period-specific fits:

$$\widehat{\mu}(\beta; \mathbf{q}, \{J_t\}) = \frac{1}{T - H} \sum_{t=H+1}^T \widehat{M}_t(\beta; \mathbf{q}, J_t).$$

Given choices of bandwidths  $h_{it}$  and portfolio counts  $J_t$ , the classical beta-sorted portfolio estimator is  $\widehat{\mu}(\beta; 0, \{J_t\})$  constructed using  $\widehat{\beta}_{it}(0, h_{it})$ . The bias-corrected estimator used for inference is  $\widehat{\mu}(\beta; 1, \{J_t\})$  constructed using  $\widehat{\beta}_{it}(1, h_{it})$ .

The bandwidths  $h_{it}$  and portfolio counts  $J_t$  are chosen once for the classical point estimation objective and then reused for inference. Thus, the empirical rule is simple: choose  $h_{it}$  and  $J_t$  as if the objective were the classical estimator with  $(\mathbf{p}, \mathbf{q}) = (0, 0)$ , but form confidence intervals with the bias-corrected estimator with  $(\mathbf{p}, \mathbf{q}) = (1, 1)$ . This is the usual robust bias-correction logic: the lower-order fit determines the MSE-optimal tuning, and the higher-order fit removes the leading smoothing bias at that same tuning; see Calonico et al. (2014) and Calonico et al. (2018, 2022).

The plug-in logic is the same in both steps:

1. run a pilot regression at the point estimation order and an auxiliary pilot regression at the higher order;
2. estimate the variance constant from the point estimation pilot;
3. estimate the leading bias constant from the difference between the two pilot fits; and
4. plug the two constants into the MSE-optimal tuning formula.

### SA-III.1.1 First-Step Bandwidth Selection

For first-step point estimation of order  $\mathbf{p} \geq 0$ , we use squared Euclidean loss for the vector  $\widehat{\beta}_{it}(\mathbf{p}, h_{it})$ . Accordingly, let  $\mathcal{V}_{\text{FS},it}(\mathbf{p})$  denote the trace of its leading conditional covariance matrix and let  $\mathcal{B}_{\text{FS},it}^2(\mathbf{p})$  denote the squared Euclidean norm of its leading bias vector. In the smooth-trend case considered in Lemma SA-II.2.4(i) with  $s \geq \mathbf{p} + 1$ , the point estimation first-step (conditional) MSE has the expansion

$$\text{MSE}(\widehat{\beta}_{it}(\mathbf{p}, h_{it})) = \mathcal{V}_{\text{FS},it}(\mathbf{p})(Th_{it})^{-1} + \mathcal{B}_{\text{FS},it}^2(\mathbf{p})h_{it}^{2(\mathbf{p}+1)} + o_{\mathbb{P}}((Th_{it})^{-1} + h_{it}^{2(\mathbf{p}+1)}).$$

The MSE-optimal bandwidth is therefore

$$h_{it}^*(\mathbf{p}) = \left( \frac{\mathcal{V}_{\text{FS},it}(\mathbf{p})}{2(\mathbf{p}+1)\mathcal{B}_{\text{FS},it}^2(\mathbf{p})} \right)^{1/(2\mathbf{p}+3)} T^{-1/(2\mathbf{p}+3)},$$

provided that  $\mathcal{B}_{\text{FS},it}^2(\mathbf{p}) > 0$ . Algorithm SA-III.1.1 implements this rule for the classical  $\beta$ -estimation underlying the beta-sorted portfolio estimator.

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#### Algorithm SA-III.1.1: First-step MSE-optimal bandwidth ( $\mathbf{p} = 0$ )

---

**Input:** Pilot bandwidth  $h_{it,\text{pilot}}$

**Pilot regressions at  $h_{it,\text{pilot}}$ :**

order  $\mathbf{p} = 0$  fit  $\widehat{\beta}_{it}(0, h_{it,\text{pilot}})$  of  $R_{i,t-s}$  on  $X_s(t)$  over  $s \in [1, \lceil Th_{it,\text{pilot}} \rceil]$

order  $\mathbf{p} = 1$  fit  $\widehat{\beta}_{it}(1, h_{it,\text{pilot}})$  over the same window and with the same kernel

**Variance estimate:**  $\widehat{\mathcal{V}}_{\text{FS},it}(0) \leftarrow$  trace of the sandwich covariance constant from the  $\mathbf{p} = 0$  pilot

**Squared-bias estimate:**  $\widehat{\mathcal{B}}_{\text{FS},it}^2(0) \leftarrow \left\| (\widehat{\beta}_{it}(1, h_{it,\text{pilot}}) - \widehat{\beta}_{it}(0, h_{it,\text{pilot}})) / h_{it,\text{pilot}} \right\|^2$

**Plug-in bandwidth:**

$$\widehat{h}_{it}^* \leftarrow \left( \frac{\widehat{\mathcal{V}}_{\text{FS},it}(0)}{2\widehat{\mathcal{B}}_{\text{FS},it}^2(0)} \right)^{1/3} T^{-1/3}$$

**Estimates:** compute  $\widehat{\beta}_{it} = \widehat{\beta}_{it}(0, \widehat{h}_{it}^*)$  and  $\widehat{\beta}_{it}^{\text{BC}} = \widehat{\beta}_{it}(1, \widehat{h}_{it}^*)$

**Output:**  $\widehat{h}_{it}^*$ ,  $\widehat{\beta}_{it}$ , and  $\widehat{\beta}_{it}^{\text{BC}}$ .

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### SA-III.1.2 Second-Step Selection of the Number of Portfolios

For second-step point estimation using within-portfolio polynomials of order  $\mathbf{q} \geq 0$ , we use the same marginal portfolio count  $J_t$  in every beta coordinate. Thus  $J_t$  is a single scalar selected for period  $t$ , and the tensor-product partition contains at most  $J_t^d$  cells. Let  $\mathcal{V}_{\text{SS},t}(\beta; \mathbf{q})$  denote the leading variance constant of  $\widehat{M}_t(\beta; \mathbf{q}, J_t)$ , and let  $\mathcal{B}_{\text{SS},t}(\beta; \mathbf{q})$  denote the leading approximation-bias constant multiplying  $J_t^{-(\mathbf{q}+1)}$ . Up to additive terms that do not depend on the portfolio count, including the factor time-series component present when  $\beta \neq 0$ , we use the period- $t$  point estimation approximate MSE criterion

$$\text{MSE}_t(\widehat{M}_t(\beta; \mathbf{q}, J_t)) = \mathcal{V}_{\text{SS},t}(\beta; \mathbf{q}) \frac{J_t^d}{n_t} + \mathcal{B}_{\text{SS},t}(\beta; \mathbf{q})^2 J_t^{-2(\mathbf{q}+1)} + o_{\mathbb{P}}(J_t^d/n_t + J_t^{-2(\mathbf{q}+1)}).$$

To obtain one portfolio count for the whole beta domain, fix a nonnegative weight function  $w(\beta)$  on  $\mathcal{B}^\circ$

satisfying  $0 < \int_{\mathcal{B}^\circ} w(\beta) d\beta < \infty$ , and define the integrated variance and squared-bias constants

$$\begin{aligned}\mathcal{V}_{\text{SS},t}^w(\mathbf{q}) &:= \int_{\mathcal{B}^\circ} \mathcal{V}_{\text{SS},t}(\beta; \mathbf{q}) w(\beta) d\beta, \\ \mathcal{B}_{\text{SS},t}^{2,w}(\mathbf{q}) &:= \int_{\mathcal{B}^\circ} \mathcal{B}_{\text{SS},t}(\beta; \mathbf{q})^2 w(\beta) d\beta.\end{aligned}$$

The corresponding approximate integrated MSE is

$$\begin{aligned}\text{IMSE}_t(\mathbf{q}, J_t; w) &= \mathcal{V}_{\text{SS},t}^w(\mathbf{q}) \frac{J_t^d}{n_t} + \mathcal{B}_{\text{SS},t}^{2,w}(\mathbf{q}) J_t^{-2(\mathbf{q}+1)} \\ &\quad + o_{\mathbb{P}}(J_t^d/n_t + J_t^{-2(\mathbf{q}+1)}).\end{aligned}$$

Thus the IMSE-optimal common number of portfolios per beta dimension is

$$J_t^*(\mathbf{q}; w) = \left( \frac{2(\mathbf{q}+1) \mathcal{B}_{\text{SS},t}^{2,w}(\mathbf{q})}{d \mathcal{V}_{\text{SS},t}^w(\mathbf{q})} \right)^{1/(d+2\mathbf{q}+2)} n_t^{1/(d+2\mathbf{q}+2)},$$

provided that  $\mathcal{B}_{\text{SS},t}^{2,w}(\mathbf{q}) > 0$ . Algorithm [SA-III.1.2](#) implements this rule for the classical beta-sorted portfolio estimator.

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**Algorithm SA-III.1.2:** Second-step IMSE-optimal portfolios ( $\mathbf{q} = 0$ )

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**Input:** First-step point estimates  $\{\hat{\beta}_{it}\}$  from Algorithm [SA-III.1.1](#); pilot portfolio counts  $\{J_{t,\text{pilot}}\}$  with common scale  $J_{\text{pilot}}$ ; grid  $\mathcal{G} = \{\beta_\ell : \ell \in [L]\} \subset \mathcal{B}^\circ$ ; nonnegative grid weights  $\{w_\ell : \ell \in [L]\}$

**Pilot portfolio regressions at  $\{J_{t,\text{pilot}}\}$ :**

second-step order  $\mathbf{q} = 0$  portfolio regression  $\widehat{M}_t(\cdot; 0, J_{t,\text{pilot}})$

second-step order  $\mathbf{q} = 1$  portfolio regression  $\widehat{M}_t(\cdot; 1, J_{t,\text{pilot}})$

**Pointwise variance estimates:**  $\widehat{\mathcal{V}}_{\text{SS},t}(\beta_\ell; 0) \leftarrow$  period- $t$  sandwich variance constant from the  $\mathbf{q} = 0$  pilot for each  $\ell \in [L]$

**Integrated variance estimate:**  $\widehat{\mathcal{V}}_{\text{SS},t}^w(0) \leftarrow \sum_{\ell=1}^L w_\ell \widehat{\mathcal{V}}_{\text{SS},t}(\beta_\ell; 0)$

**Integrated squared-bias estimate:**

$$\widehat{\mathcal{B}}_{\text{SS},t}^{2,w}(0) \leftarrow \sum_{\ell=1}^L w_\ell \left[ \widehat{M}_t(\beta_\ell; 1, J_{t,\text{pilot}}) - \widehat{M}_t(\beta_\ell; 0, J_{t,\text{pilot}}) \right]^2 J_{t,\text{pilot}}^2$$

**Plug-in number of portfolios:**

$$\widehat{J}_t^* \leftarrow \left( \frac{2 \widehat{\mathcal{B}}_{\text{SS},t}^{2,w}(0)}{d \widehat{\mathcal{V}}_{\text{SS},t}^w(0)} \right)^{1/(d+2)} n_t^{1/(d+2)}$$

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**Output:** The common marginal portfolio count  $\widehat{J}_t^*$ .

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Our default implementation takes  $w_\ell = 1$  for every grid point. Hence the estimated squared-bias constant is the sum of the squared pointwise pilot biases and the estimated variance constant is the sum of the pointwise

pilot variance constants. Normalizing the grid weights to sum to one would leave  $\hat{J}_t^\star$  unchanged because the same normalization multiplies the numerator and denominator. On an irregular grid, the  $w_\ell$  may instead combine the desired  $w(\beta_\ell)$  with quadrature weights; this changes the discrete approximation to the integral but not the form of the selector.

### SA-III.1.3 Estimation and Robust Bias-Corrected Inference

The classical beta-sorted portfolio estimator constructed using the first- and second-step plug-in rules is

$$\hat{\mu}(\beta) = \hat{\mu}(\beta; 0, \{\hat{J}_t^\star\}),$$

which is computed using the  $\mathbf{p} = 0$  first-step estimates  $\{\hat{\beta}_{it}\}$  from Algorithm SA-III.1.1.

A robust bias-corrected (RBC) confidence interval uses the  $\mathbf{p} = \mathbf{q} = 1$  fits with the selected tuning parameters  $\{\hat{h}_{it}^\star\}$  and  $\{\hat{J}_t^\star\}$ , and the bias-corrected first-step point estimates  $\{\hat{\beta}_{it}^{\text{BC}}\}$  from Algorithm SA-III.1.1. Specifically, a  $100(1 - \alpha)\%$  nominal level RBC confidence interval is

$$\hat{\mu}^{\text{BC}}(\beta) \pm z_{1-\alpha/2} \sqrt{\hat{\Sigma}^{\text{BC}}(\beta)},$$

where  $\hat{\mu}^{\text{BC}}(\beta) = \hat{\mu}(\beta; 1, \{\hat{J}_t^\star\})$  using  $\{\hat{\beta}_{it}^{\text{BC}}\}$ ,  $\{\hat{h}_{it}^\star\}$ , and  $\{\hat{J}_t^\star\}$ , and  $\hat{\Sigma}^{\text{BC}}(\beta) = \hat{\Sigma}(\beta)$  as in Section SA-II.7 for  $\hat{\mu}^{\text{BC}}(\beta)$ . For each fixed  $\beta \neq 0$ , under the pointwise Gaussian-approximation conditions and the conditions of Theorem SA-II.7.3, this interval has exact pointwise coverage when  $T\mathcal{B}_F(\beta) = o_{\mathbb{P}}(1)$ ; without this additional factor-prediction rate it is conservative because  $\mathcal{B}_F(\beta) \geq 0$ , and no anchoring condition is needed because Lemma SA-II.7.2 gives  $|\hat{\Sigma}_\varepsilon(\beta)| = O_{\mathbb{P}}(J^d/(nT)) = o_{\mathbb{P}}(T^{-1})$ .

### SA-III.1.4 Validity of Tuning Parameter Selection

The next lemma checks the polynomial-rate restrictions for the plug-in scaling. The key distinction is the normalization. Pointwise inference at a fixed nonzero  $\beta$  is governed by the factor scale. Pointwise inference at  $\beta = 0$  and factor-cancelling contrasts are governed by the idiosyncratic scale, which magnifies the components of  $\mathbf{R}_{\text{LA}}$  that are harmless at the factor scale.

**Lemma SA-III.1.1** (Validity of Pointwise-in- $\beta$  Tuning). *Suppose  $T = n^\eta$  for some  $\eta > 0$ , and suppose that Assumptions SA-DGP, SA-FS, SA-PA, and SA-FSB hold and that  $s_\mu \geq 2$ . Suppose the plug-in choices satisfy, uniformly over their indices,*

$$\hat{h}_{it}^\star \asymp h \asymp T^{-1/3}, \quad \hat{J}_t^\star \asymp J \asymp n^{1/(d+2)},$$

*and take the moment exponent in Assumption SA-DGP sufficiently large. Then the following rate implications hold.*

- (i) Pointwise inference for  $\beta \neq 0$ . *The plug-in scaling verifies the factor-scale restrictions for fixed-nonzero- $\beta$  inference if*

$$0 < \eta < \frac{4}{d+2}. \tag{SA-III.1.1}$$

In addition, the deterministic threshold in Assumption [SA-FSB](#) must satisfy

$$\sqrt{T} \mathbf{B}_{\text{FS}} = o(1).$$

The upper bound is the factor-scale second-step smoothing-bias condition for the  $\mathbf{q} = 1$  inference fit,

$$\sqrt{T} J^{-2} \rightarrow 0,$$

which is equivalent to  $\eta < 4/(d+2)$  under  $J \asymp n^{1/(d+2)}$  and  $T = n^\eta$ .

- (ii) Pointwise inference for  $\beta = 0$  and factor-cancelling contrasts. Pointwise inference at  $\beta = 0$  in Theorem [SA-II.6.2](#) and factor-cancelling contrast inference in Theorem [SA-II.9.1](#) both require the linearization error to be negligible after normalization by  $\sqrt{nT}/J^d$ . Ignoring logarithmic factors, the third component of  $\mathbf{R}_{\text{LA}}$  in Lemma [SA-II.4.2](#) requires

$$\sqrt{\frac{nT}{J^d}} \sqrt{\frac{J^{2d}}{nHJ^{d-1} \max\{J, \sqrt{H}\}}} = \sqrt{\frac{TJ}{H \max\{J, \sqrt{H}\}}} \rightarrow 0. \quad (\text{SA-III.1.2})$$

With  $H \asymp T^{2/3}$  and  $J \asymp n^{1/(d+2)}$ , the right-hand side is equal to

$$\begin{cases} T^{1/6}, & J \geq \sqrt{H}, \\ J^{1/2}, & J < \sqrt{H}, \end{cases}$$

and therefore diverges. Thus the plug-in scaling cannot verify the natural origin-scale RBC rate restrictions. In addition, the second-step smoothing-bias condition alone would require

$$\frac{nT}{J^{d+4}} \rightarrow 0 \quad \Longleftrightarrow \quad \eta < \frac{2}{d+2},$$

while the deterministic  $H^{-1}$  component of  $\mathbf{R}_{\text{LA}}$  would require, up to logarithmic factors,

$$\eta > \frac{6}{d+2}.$$

The first-step bias component would separately require

$$\sqrt{\frac{nT}{J^d}} \mathbf{B}_{\text{FS}} = o(1).$$

The stronger primitive condition  $J\mathbf{R}_{\text{FS}} = o(1)$  is not needed for this fixed-nonzero- $\beta$  factor-scale calculation; it is an idiosyncratic-scale sufficient condition for absorbing the basis-switching term into the local-support scale. Thus, the recommended choices of  $h$  and  $J$ , together with  $\sqrt{T} \mathbf{B}_{\text{FS}} = o(1)$ , verify fixed-nonzero- $\beta$  pointwise inference within the growth window in [\(SA-III.1.1\)](#), but do not verify idiosyncratic-scale RBC inference at the origin or for factor-cancelling contrasts.

### SA-III.1.5 Proof of Lemma SA-III.1.1

Throughout the proof, exponent comparisons suppress logarithmic factors. The additional first-step bias restriction in part (i) gives  $\sqrt{T} B_{\text{FS}} = o(1)$ . The first-step variance terms satisfy

$$\sqrt{T} \frac{1}{Th} \asymp T^{-1/6} \rightarrow 0,$$

and the remaining first-step terms involving the moment exponent are negligible by taking that exponent sufficiently large. For the second-step portfolio fit with  $q = 1$  used for inference, the smoothing bias is of order  $J^{-2}$ . Since  $J \asymp n^{1/(d+2)}$  and  $T = n^\eta$ ,

$$\sqrt{T} J^{-2} \rightarrow 0 \iff n^{\eta/2-2/(d+2)} \rightarrow 0 \iff \eta < \frac{4}{d+2},$$

which gives the upper bound in (SA-III.1.1). The main portfolio variance restriction is  $J^d/(nh) = o(1)$ . With the same substitutions,

$$\frac{J^d}{nh} \asymp n^{d/(d+2)-1+\eta/3},$$

so this term is  $o(1)$  whenever  $\eta < 6/(d+2)$ , which is implied by (SA-III.1.1). This condition also implies  $J^d/n = o(1)$ , which is the effective nonempty-portfolio requirement in the consistency and variance-estimation arguments. The sharpened jackknife restriction for the  $q = 1$  inference fit requires  $HJ^{-4} = o(1)$  up to logarithmic factors. Since

$$HJ^{-4} \asymp n^{2\eta/3-4/(d+2)},$$

this condition follows from (SA-III.1.1). The remaining terms in (SA-II.7.1) are smaller under the same condition and the first-step rates above. The endpoint condition also holds, since

$$\frac{J^2}{n} \asymp n^{2/(d+2)-1} = o(1).$$

The full linear-approximation remainder is negligible at the factor scale. The deterministic pieces satisfy

$$\sqrt{T} \left( \frac{1}{H} + B_{\text{FS}} \right) \lesssim T^{-1/6} + \sqrt{T} B_{\text{FS}} = o(1).$$

The empirical-process component in  $R_{\text{LA}}$  satisfies

$$\sqrt{T} \sqrt{\frac{J^{2d}}{nH J^{d-1} \max\{J, \sqrt{H}\}}} = \begin{cases} \sqrt{T J^d/(nH)}, & J \geq \sqrt{H}, \\ \sqrt{T J^{d+1}/(nH^{3/2})}, & J < \sqrt{H}, \end{cases} = o(1),$$

where the first case uses  $\eta < 6/(d+2)$  and the second case is  $n^{-1/(2(d+2))}$ . The Gram-perturbation term is also negligible. Indeed, Lemma SA-II.3.3 gives  $J^d R_q = o(1)$  and

$$\sqrt{T} J^d R_q \sqrt{\frac{J^d}{nH}} \leq J^d R_q \sqrt{\frac{J^d}{nh}} = o(1),$$

where logarithmic factors are suppressed. The basis-switching term is also negligible at the factor scale without imposing the stronger idiosyncratic-scale condition  $J^d \mathbf{R}_P = o(1)$ . Since the minimum operators in  $\mathbf{R}_P$  are bounded by one,

$$\mathbf{R}_P \lesssim J^{-d} + \sqrt{\frac{J^{-d} \log(nT)}{n}} + \frac{\log(nT)}{n}.$$

Therefore Lemma [SA-I.3.1](#) gives

$$\sqrt{T} |\mathcal{R}_2(\beta)| \lesssim_{\mathbb{P}} \sqrt{\frac{J^{2d} \mathbf{R}_P}{n}} = o_{\mathbb{P}}(1),$$

because  $J \asymp n^{1/(d+2)}$  implies  $J^d/n = o(1)$ . The same-cell nonconstant-basis component in  $\mathbf{R}_{\text{LA}}$  is negligible away from the origin. At the plug-in scaling, for fixed  $\beta \neq 0$ ,

$$\sqrt{T} \sqrt{\frac{J^{d+2}}{nH^2}} \asymp T^{-1/6} = o(1),$$

because  $J^{d+2} \asymp n$  and  $H \asymp T^{2/3}$ . The endpoint-interior term is smaller:

$$\sqrt{T} \sqrt{\frac{J^{d+2}}{n^2 H}} \asymp \sqrt{\frac{T J^{d+2}}{n^2 H}} = \sqrt{\frac{T}{nH}} = o(1).$$

Thus the infeasible Gaussian approximation holds for each fixed  $\beta \neq 0$  at the plug-in scaling.

At the origin, the relevant normalization is  $\sqrt{nT/J^d}$ . The third component of  $\mathbf{R}_{\text{LA}}$  gives the necessary condition in [\(SA-III.1.2\)](#). At the plug-in scaling,

$$\sqrt{\frac{nT}{J^d}} \sqrt{\frac{J^{2d}}{nH J^{d-1} \max\{J, \sqrt{H}\}}} = \begin{cases} T^{1/6}, & J \geq \sqrt{H}, \\ J^{1/2}, & J < \sqrt{H}, \end{cases}$$

which diverges. Hence  $\mathbf{R}_{\text{CLT}}/\sqrt{\Sigma(0)}$  cannot vanish under the plug-in scaling. Even ignoring this obstruction, the smoothing-bias component in  $\mathbf{R}_{\text{CLT}}$  would require

$$\sqrt{\frac{nT}{J^d}} J^{-2} \rightarrow 0 \iff \frac{nT}{J^{d+4}} \rightarrow 0 \iff \eta < \frac{2}{d+2}.$$

The deterministic and smooth-interior components of  $\mathbf{R}_{\text{LA}}$  also give

$$\sqrt{\frac{nT}{J^d}} \left( \frac{1}{H} + \mathbf{B}_{\text{FS}} + \sqrt{\frac{J^{d+2}}{nH^2}} \right) \gtrsim \frac{J\sqrt{T}}{H},$$

up to logarithmic factors. Under the plug-in scaling,

$$\frac{J\sqrt{T}}{H} \asymp n^{1/(d+2)-\eta/6},$$

so these components would require  $\eta > 6/(d+2)$ , which is incompatible with the smoothing-bias requirement

above. The first-step bias component would additionally require

$$\sqrt{\frac{nT}{J^d}} B_{\text{FS}} = o(1).$$

The recommended tuning therefore does not verify natural origin-scale RBC inference. The same obstruction applies to factor-cancelling contrasts. In Theorem SA-II.9.1, the factor term cancels, so the condition (SA-II.9.3) normalizes the residual at the idiosyncratic scale  $\sqrt{nT/J^d}$ . By definition,  $R_\Psi = (\sum_{k=1}^{K_\Psi} |a_k|) R_{\text{CLT}}$ , and therefore it contains the same  $R_{\text{LA}}$  component. Thus (SA-II.9.3) imposes the same necessary condition (SA-III.1.2), which diverges under the plug-in scaling.  $\square$

## SA-III.2 Uniform-in- $\beta$

This section gives the tuning and implementation steps for the uniform-in- $\beta$  confidence intervals of Theorem SA-II.8.1. Throughout, “uniform-in- $\beta$ ” means that the coverage probability is controlled uniformly over evaluation points  $\beta \in \mathcal{B}^\circ$ , in the pointwise-uniform sense defined in Section SA-II.8; it is not a simultaneous confidence band for the full function  $\mu(\cdot)$ . The region  $\mathcal{B}^\circ$  is the same fixed interior region used in Theorem SA-II.8.1, and it contains a fixed neighborhood of the origin.

We keep the first-step bandwidths and second-step portfolio counts selected in Section SA-III.1. The rate calculation uses the same uniform orders as in Lemma SA-III.1.1, namely  $\hat{h}_{it}^* \asymp T^{-1/3}$  and  $\hat{J}_t^* \asymp n_t^{1/(d+2)}$ . Algorithm SA-III.1.2 implements a grid approximation to the period-specific weighted IMSE criterion.

We separate the final time-average burn-in from the selected local first-step windows. Let  $T_0$  denote a pre-specified implementation burn-in, chosen as a deterministic cap for the selected first-step windows so that, with probability approaching one,

$$T_0 \geq \max_{1 \leq s \leq T} \max_{i \in [n_s]} \hat{H}_{is}^*, \quad \hat{H}_{is}^* = \lceil T \hat{h}_{is}^* \rceil,$$

and  $T_0 \asymp T^{2/3}$ . The final time-series averages below use  $t = T_0 + 1, \dots, T$ , whereas each first-step local-polynomial fit still uses its own selected window  $\hat{H}_{it}^*$ . The remaining tuning parameter is the near-origin truncation parameter  $\delta$ , which determines where the usual feasible standard error is replaced by the conservative variance floor from Theorem SA-II.8.1.

### SA-III.2.1 Near-Origin Truncation Parameter

Theorem SA-II.8.1 uses a scalar threshold  $\delta$  in factor-variance units. Its role in the implementation is only to identify a small neighborhood of the origin in which the usual factor-based standard error is replaced by a conservative floor. We therefore use a deliberately simple rule of thumb rather than attempting to estimate every remainder constant.

For the degree-one second step used in the robust bias-corrected implementation, we choose a threshold of the form

$$\hat{\delta} := c_\delta \hat{s}_R^2 (\log T)^{d+1} (\log \log T) \max \{ \hat{r}_J, \hat{r}_H, \hat{r}_N \}. \quad (\text{SA-III.2.1})$$

Here  $\hat{r}_J$ ,  $\hat{r}_H$ , and  $\hat{r}_N$  estimate the rate components associated with the squared degree-one second-step approximation bias, the squared local-window remainder, and the uniform first-step stochastic effect, re-

spectively;  $\hat{s}_R^2$  puts the threshold in return- variance units. Section SA-III.4 gives the exact finite-sample definitions of these quantities and the default choice of  $c_\delta$ . The adjustment factor  $(\log T)^{d+1}(\log \log T)$  ensures that the threshold rate restrictions required by Theorem SA-II.8.1 are satisfied under our recommended tuning parameter choices, provided that  $B_{FS} = O(H^{-1})$ .

Under independent factors and homoskedastic first-step errors, converting coefficient noise back to factor-variance units cancels the factor variance; local portfolio averaging then gives the rate  $J^d/(nH)$  underlying  $\hat{r}_N$ .

The rule is scale equivariant in the intended sense. Multiplying returns by  $a$  multiplies  $\hat{s}_R^2$  and  $\hat{\delta}$  by  $a^2$ . A diagonal rescaling of a beta coordinate paired with the inverse rescaling of its factor leaves the switching comparison unchanged. Coordinate permutations leave the rate candidates unchanged. The rule is a finite-sample calibration device, not a claim that the unknown optimal threshold constant has been consistently estimated. Formal use of Theorem SA-II.8.1 still requires the resulting scalar threshold to satisfy (SA-II.8.1).

### SA-III.2.2 Estimation and Robust Bias-Corrected Inference

Given  $\hat{h}_{it}^*$ , the common portfolio-count scale  $\hat{J}_t^*$ , and the threshold  $\hat{\delta}$ , compute the same robust bias-corrected estimator as in Subsection SA-III.1.3, but with the common uniform-in- $\beta$  portfolio-count sequence and the final averaging range  $t = T_0 + 1, \dots, T$ :

$$\hat{\mu}^{BC}(\beta) = \frac{1}{T - T_0} \sum_{t=T_0+1}^T \widehat{M}_t(\beta; 1, \hat{J}_t^*),$$

using the bias-corrected first-step estimates  $\{\hat{\beta}_{it}^{BC}\}$  and the selected bandwidths  $\{\hat{h}_{it}^*\}$ . For this estimator, compute the feasible factor component  $\hat{\Sigma}_f(\beta)$  and the idiosyncratic component  $\hat{\Sigma}_\varepsilon^{BC}(\beta)$  as in Section SA-II.7, but over the same final averaging range  $t = T_0 + 1, \dots, T$ , with the idiosyncratic component using the residuals and weights from the same robust bias-corrected fit. Set

$$\hat{\Sigma}^{BC}(\beta) = \hat{\Sigma}_f(\beta) + \hat{\Sigma}_\varepsilon^{BC}(\beta).$$

The conservative variance floor is

$$\hat{\Sigma}_\delta^{BC} = \sup_{\substack{b \in \mathcal{B}^\circ \\ \hat{\Sigma}_f(b) = \hat{\delta}}} \max\{\hat{\Sigma}^{BC}(b), \hat{\delta}\} = \hat{\delta} + \sup_{\substack{b \in \mathcal{B}^\circ \\ \hat{\Sigma}_f(b) < \hat{\delta}}} \max\{\hat{\Sigma}_\varepsilon^{BC}(b), 0\}.$$

If the level set is empty in a finite sample, set  $\hat{\Sigma}_\delta^{BC} = \hat{\delta}$ . For numerical work, the boundary supremum is evaluated on the grid  $\mathcal{G}$ , using interpolation or the nearest grid points to the level set  $\hat{\Sigma}_f(b) = \hat{\delta}$ . This grid approximation is purely computational; the theoretical statement above is the continuum version.

The variance used in the uniform-in- $\beta$  robust bias-corrected interval is

$$\hat{\Sigma}_{UI}^{BC}(\beta) = \begin{cases} \hat{\Sigma}^{BC}(\beta) & \text{if } \hat{\Sigma}_f(\beta) > \hat{\delta} \\ \max\{\hat{\Sigma}^{BC}(\beta), \hat{\Sigma}_\delta^{BC}\} & \text{if } \hat{\Sigma}_f(\beta) \leq \hat{\delta} \end{cases},$$

The resulting  $100(1 - \alpha)\%$  interval is

$$\hat{\mu}^{BC}(\beta) \pm z_{1-\alpha/2} \sqrt{\hat{\Sigma}_{UI}^{BC}(\beta)}.$$

### SA-III.3 Factor-Cancelling Linear Contrast

Factor-cancelling contrasts do not introduce a new threshold parameter. Their distinctive feature is the normalization: after the factor component cancels, Theorem SA-II.9.1 works at the idiosyncratic scale  $\sqrt{nT/J^d}$ . Thus a factor-cancelling contrast is covered by the theory only when the selected first-step bandwidths and second-step portfolio counts verify

$$\sqrt{\frac{nT}{J^d}} R_\Psi \rightarrow 0,$$

together with the remaining conditions in Theorem SA-II.9.1. For the fixed finite contrast, the definition above yields

$$R_\Psi = \left( \sum_{k=1}^{K_\Psi} |a_k| \right) R_{\text{CLT}},$$

so this condition is equivalent to imposing the same normalized rate on  $R_{\text{CLT}}$ .

The pointwise first-step and IMSE second-step plug-in choices from Algorithm SA-III.1.1 and Algorithm SA-III.1.2 are therefore not a validated default for factor-cancelling inference. Lemma SA-III.1.1 shows that, under those choices, the idiosyncratic-scale normalization magnifies the linear-approximation remainder in  $R_{\text{LA}}$ , and the required rate diverges. Consequently, a factor-cancelling confidence interval should use the jackknife variance estimator in Theorem SA-II.9.1 only with a dedicated undersmoothing choice of  $h$  and  $J$  that satisfies the display above. Operationally, this means choosing  $h$  and  $J$  to make the estimated non-leading remainder small relative to  $\sqrt{J^d/(nT)}$ , rather than choosing  $J$  to minimize the IMSE of the lower-order estimator.

### SA-III.4 Default Implementation Choices

In this section, we describe the implementation conventions used in the two empirical applications and the Monte Carlo simulations.

**Tuning parameters** We follow the approach described in Section SA-III.1. The pilot choices are  $H_{\text{pilot}} = 10$  years (120 months or 40 quarters) and  $J_{\text{pilot}} = 10$ . Thus,  $h_{\text{pilot}} = 120/T$  for monthly data and  $h_{\text{pilot}} = 40/T$  for quarterly data. Here  $h = H/T$ , so the denominator is the full time-series sample size  $T$ , rather than the  $T - H$  dates entering the second-step average. We make a number of finite-sample regularizations to ensure that the first-step and second-step estimators are well behaved. For the first step we restrict the selected window to  $36 \leq H_{it} \leq 360$  for monthly data and  $36 \leq H_{it} \leq 120$  for quarterly data. For an unbalanced panel, we require at least 36 usable observations within the selected window to estimate a beta. In addition, for inference, we use  $\mathbf{p} = 1$  so long as  $\widehat{\text{Var}}_{\text{HC1}}(\hat{\beta}_{it}^{(1)}) \leq T^{1/10} \widehat{\text{Var}}_{\text{HC1}}(\hat{\beta}_{it}^{(0)})$ , and use  $\mathbf{p} = 0$  otherwise. Because  $T \rightarrow \infty$ , this regularization becomes asymptotically inactive under the conditions in SA-Theory. Similarly, if the  $\mathbf{p} = 1$  design matrix is singular, we use the  $\mathbf{p} = 0$  estimate instead. We constrain the number of portfolios to  $2 \leq J_t \leq 40$ . The lower bound ensures the data are split into separate portfolios whereas the upper bound helps prevent the formation of portfolios containing too few assets. The grids used to select  $J_t$  and to report estimates and inference results are chosen separately. In each eligible post-burn-in period, the implementation finds the largest origin-centered interval contained in the cross-sectional range of

the relevant estimated betas. It then uses the largest such symmetric interval that is contained in the realized range in at least 90% of the eligible periods, weighting periods equally. The  $J_t$ -selection grid consists of 25 equally spaced points on this interval and is constructed from the point-estimation betas. The reporting and inference grid consists of 11 equally spaced points and is constructed from the first-step beta estimates used for inference. Hence the reporting grid contains zero exactly but does not affect the choice of  $J_t$ . For graphical clarity, the business-credit application uses a manually specified reporting grid from  $-1.25$  to  $1.25$  in increments of  $0.25$ , together with a separate grid of 25 equally spaced points on  $[-1, 1]$  for selecting  $J_t$ . For simplicity, we use the round endpoints  $-1$  and  $1$ , which are close to those selected by the default support rule. This interval lies within the realized range of the point-estimation betas in 85.7% of months, compared with the default target of 90%.

**Variance estimation** For the common-factor component of the variance estimator, both empirical applications forecast the risk factor using an AR(1) with an intercept, estimated recursively by OLS using only prior observations. Since the first-step window  $H_{it}$  generally varies across assets and dates, the implementation uses a pair-specific lag cutoff in the JN variance estimator. For observations on asset  $i$  at dates  $t$  and  $s$ , it retains the jackknife cross-product whenever  $|t - s| \leq \max\{H_{it}, H_{is}\}$ . This rule accommodates the dependence induced by overlapping first-step windows of different lengths without applying the largest selected window to every asset-date pair. If  $H_{it} = H$  for all  $i$  and  $t$ , it reduces to the common lag- $H$  truncation in equation (4.11) of the main text.

**Uniform-in- $\beta$  inference** For each post-burn-in date  $t$ , let  $\mathcal{I}_t^\delta$  denote the set of assets included in the second-step estimator used for inference. Let  $n_t^\delta = |\mathcal{I}_t^\delta|$  and  $\mathcal{I}^\delta = \{(i, t) : i \in \mathcal{I}_t^\delta\}$ . Because returns can contain extreme observations, we use the square of the pooled median absolute deviation as a robust estimate of the return variance scale, rescaled to be consistent for the variance under Gaussianity:

$$\hat{s}_{R, \text{MAD}}^2 = \left\{ 1.4826 \operatorname{median}_{(i, t) \in \mathcal{I}^\delta} \left| R_{it} - \operatorname{median}_{(j, s) \in \mathcal{I}^\delta} R_{js} \right| \right\}^2.$$

We compute the three rate components as observation-level averages:

$$\hat{r}_J = \frac{1}{|\mathcal{I}^\delta|} \sum_{(i, t) \in \mathcal{I}^\delta} J_t^{-4}, \quad \hat{r}_H = \frac{1}{|\mathcal{I}^\delta|} \sum_{(i, t) \in \mathcal{I}^\delta} H_{it}^{-2}, \quad \hat{r}_N = \frac{1}{|\mathcal{I}^\delta|} \sum_{(i, t) \in \mathcal{I}^\delta} \frac{J_t^d}{n_t^\delta H_{it}}.$$

Finally, we choose  $c_\delta = 1/160$  so that

$$\hat{\delta} = \frac{1}{160} \hat{s}_{R, \text{MAD}}^2 (\log T)^{d+1} (\log \log T) \max\{\hat{r}_J, \hat{r}_H, \hat{r}_N\}.$$

Each included observation receives equal weight, so periods with larger effective cross sections receive proportionally more weight in these pooled averages. The calibration sample is determined before evaluating the reporting grid; consequently, changing only the reporting grid does not alter  $\hat{\delta}$ .

**Affine Specification Test** For the business-credit application, we implement the affine specification test described in Section 5.2 of the main text, as follows. We estimate both betas using local quadratic regressions,  $p = 2$ , over two adjacent, nonoverlapping windows of  $H = 63$  months each. We choose  $H = 63$  as it is the time-series median of the cross-sectional median series shown in Figure SA-IV.2.1. We continue to require

at least 36 nonmissing observations in each window. At each date, we regress returns on an intercept and the most-recent beta by using the earlier beta as the instrument. We then form equal-weighted portfolios of the resulting residuals by sorting on the earlier beta, with  $q = 0$ . We select  $J_t$  on the matched sample using the MSE criterion with earlier local-constant beta estimates and bounds of 2 and 40 portfolios. The testing grid consists of the seven points from  $-0.75$  to  $0.75$  in increments of  $0.25$ . The covariance estimator retains cross-products of the same firm's weighted residual contributions for outcome dates at most 63 calendar months apart, accounting for overlap between recent estimation windows and outcome dates. The test sample comprises 534 months, from July 1975 through December 2019. The resulting Wald statistic is 7.871; comparison with a chi-squared distribution with seven degrees of freedom gives a  $p$ -value of 0.344.

## Part IV: Numerical Evidence

### SA-IV.1 Simulations

#### SA-IV.1.1 Data-Generating Process

We consider two specifications for the risk factor and one volatility state. Let  $k \in \{1, 2\}$  index the risk factor specification, with  $\rho_1 = 0.5$  and  $\rho_2 = 0.9$ . For each  $k$ , the risk factor follows the stationary Gaussian AR(1)

$$f_{kt} = \rho_k f_{k,t-1} + u_{kt}, \quad u_{kt} \stackrel{\text{iid}}{\sim} N(0, \sigma_{u,k}^2). \quad (\text{SA-IV.1.1})$$

We fix the stationary factor standard deviation at  $\sigma_f = 0.04$  in both specifications and therefore set  $\sigma_{u,k}^2 = (1 - \rho_k^2)\sigma_f^2$ . For each specification, we draw the initial condition from  $N(0, \sigma_f^2)$ .

The volatility state follows

$$z_t = \rho_z z_{t-1} + \sqrt{1 - \rho_z^2} v_t, \quad v_t \stackrel{\text{iid}}{\sim} N(0, 1), \quad (\text{SA-IV.1.2})$$

where  $\rho_z = 0.975$ . We initialize  $z_t$  from its stationary  $N(0, 1)$  distribution. The  $z_t$  process is independent of the risk-factor process.

The return equation under factor specification  $k$  is

$$R_{it}^{(k)} = \mu_F(\beta_{it}) + \beta_{it} f_{kt} + \varsigma_{it}^{(k)} \eta_{it}. \quad (\text{SA-IV.1.3})$$

We specify the function  $\mu_F(\cdot)$ , the paths of  $\beta_{it}$ , and the volatility scale  $\varsigma_{it}^{(k)}$  below.

**Functional form of  $\mu_F(\cdot)$**  For  $\beta \in [-1, 2]$ , define the normalized exposure  $\tilde{\beta} = (\beta + 1)/3 \in [0, 1]$  and let

$$\mu_F(\beta) = -\frac{7}{200} + \frac{1}{20} \left\{ 6\tilde{\beta}^5 - 15\tilde{\beta}^4 + 10\tilde{\beta}^3 \right\}.$$

Thus, for all eight designs, the conditional expected return is

$$\mathbb{E} \left[ R_{it}^{(k)} \mid \mathcal{F}_{t-1}, \beta_{it} = \beta \right] = \mu_F(\beta) + \beta \rho_k f_{k,t-1}. \quad (\text{SA-IV.1.4})$$

Conditional expected returns therefore vary with the lagged value of the risk factor even though the structural expected-return curve  $\mu_F(\cdot)$  has no additional time-varying component.

**Paths of  $\beta_{it}$**  We consider two specifications for the beta paths. In the time-varying specification, we generate smooth paths using the low-frequency components of a persistent time-series process (see [Crump and Gospodinov, 2022](#)):

1. Constant betas:  $\beta_{it} = \beta_i$  for  $t = 1, \dots, T$ .

2. Time-varying betas:

$$\begin{aligned}\beta_{it} &= \beta_i + \sum_{j=1}^{K_\beta} a_{ij} \varphi_{j,t}, & \varphi_{j,t} &= \sqrt{\frac{2}{T}} \cos\left(\frac{t(j - \frac{1}{2})\pi}{T}\right), \\ a_{ij} &= \sqrt{\lambda_j} \xi_{ij}, & \lambda_j &= \frac{v_\beta(1 - \rho_\beta^2)}{1 + \rho_\beta^2 - 2\rho_\beta \cos(\omega_j)}, \quad \omega_j = \frac{(j - \frac{1}{2})\pi}{T}.\end{aligned}$$

Here  $a_{ij}$  and  $\xi_{ij}$  are scalar.

We set  $\mathcal{B} = [-1, 2]$  and draw  $\beta_i \stackrel{\text{iid}}{\sim} \text{Unif}[-1, 2]$ . For the time-varying beta specification, we set  $K_\beta = 5$ ,  $\rho_\beta = 0.95$ ,  $v_\beta = 0.04$ , and  $\xi_{ij} \stackrel{\text{iid}}{\sim} N(0, 1)$ . After drawing the low-frequency coefficients, for each unit we scale the time-varying component by the smaller of one and 99.9% of the largest factor that keeps the entire resulting beta path in  $\mathcal{B}$ .

**Functional form of  $\varsigma_{it}^{(k)}$  and  $\eta_{it}$**  We assume that  $\eta_{it} \stackrel{\text{iid}}{\sim} N(0, 1)$  for  $i = 1, \dots, n$  and  $t = 1, \dots, T$ . We consider two volatility specifications:

1. Constant volatility:  $(\varsigma_{it}^{(k)})^2 = (1/50)^2$ .

2. State-dependent volatility:

$$(\varsigma_{it}^{(k)})^2 = \left(\frac{1}{50}\right)^2 \exp\left\{\frac{1}{2}z_{t-1} - \frac{1}{8}\right\}.$$

Because  $z_t \sim N(0, 1)$  in the stationary distribution,  $\mathbb{E}[(\varsigma_{it}^{(k)})^2] = (1/50)^2$  under both specifications.

Within each design, the risk-factor process, volatility-state process, beta paths, and idiosyncratic return shocks are mutually independent. The complete set of experiments crosses the two persistence choices, two types of beta paths, and two volatility specifications, for a total of eight data-generating processes (DGPs) summarized in Table [SA-IV.1.1](#).

| Design | $\rho_k$ | Beta path    | Idiosyncratic volatility |
|--------|----------|--------------|--------------------------|
| S1     | 0.5      | Constant     | Constant                 |
| S2     | 0.5      | Constant     | State-dependent          |
| S3     | 0.5      | Time-varying | Constant                 |
| S4     | 0.5      | Time-varying | State-dependent          |
| S5     | 0.9      | Constant     | Constant                 |
| S6     | 0.9      | Constant     | State-dependent          |
| S7     | 0.9      | Time-varying | Constant                 |
| S8     | 0.9      | Time-varying | State-dependent          |

Table SA-IV.1.1: Simulation designs. All other calibration choices are common across designs; the factor-innovation variance varies with  $\rho_k$  as described above to hold the unconditional risk-factor variance fixed.

### SA-IV.1.2 Monte Carlo Implementation

We generate  $S = 1,000$  Monte Carlo replications for each of the eight DGPs. Each replication is a panel dataset with  $n = 5,000$  and  $T = 800$ . The first 240 months are excluded from second-step estimation and inference as a burn-in period. In each replication, we select the first-step windows  $H_{it}$  and the second-step number of portfolios  $J_t$  using the procedure described in Section SA-III.1. Apart from the longer 240-month burn-in and the evaluation grid specified below, we use the default implementation for monthly data described in Section SA-III.4.

For the factor component of the proposed covariance estimator, the implementation recursively fits via OLS the univariate AR(1)

$$f_{k\tau} = a_{k,t} + \rho_{k,t} f_{k,\tau-1} + e_{k\tau}, \quad \tau < t,$$

using only the history of the selected factor  $f_k$ , and uses the resulting one-step residual at date  $t$ . The implementation estimates the intercept and persistence recursively and does not use the true value of  $\rho_k$ .

We evaluate the SACER at

$$\mathcal{B}_{\text{eval}} = \left\{ -1 + \frac{\ell}{7} : \ell = 1, \dots, 20 \right\}.$$

For replication  $s$  and evaluation point  $\beta$ , the corresponding inferential target is

$$\bar{\mu}_s^{(k)}(\beta) = \frac{1}{T - 240} \sum_{t=241}^T \left\{ \mu_F(\beta) + \beta \rho_k f_{k,t-1}^{(s)} \right\}.$$

Since our focus is on inference, our primary estimators are the infeasible oracle with known  $\beta_{it}$  and  $\mathbf{q} = 1$ , and the feasible estimator with  $(\mathbf{p}, \mathbf{q}) = (1, 1)$ . For pointwise inference, we report the following procedures:

1. the proposed JN variance estimator;
2. the FM variance estimator formed from the  $(\mathbf{p}, \mathbf{q}) = (0, 0)$  estimates; and
3. the FM variance estimator formed from the  $(\mathbf{p}, \mathbf{q}) = (1, 1)$  estimates.

For the proposed JN variance estimator, we report the oracle and feasible estimators separately. For every reported procedure, we record pointwise coverage and average confidence-interval length.

All reported confidence intervals have nominal coverage of 95%. For the feasible estimator paired with the proposed JN variance estimator, we also report pointwise confidence intervals that are uniformly valid over  $\beta$ , as developed in Section SA-III.2. We choose  $\hat{\delta}$  using the default implementation described in Section SA-III.4. We record coverage and average interval length at every evaluation point.

### SA-IV.1.3 Results

Figure SA-IV.1.1 reports empirical coverage for the proposed pointwise and uniform-in- $\beta$  confidence intervals, together with the infeasible oracle procedure. The oracle procedure—which treats the  $\beta_{it}$  as observed—performs well across all eight designs, with average pointwise coverage across the evaluation grid ranging from 93.9% to 96.0% (blue solid line).<sup>1</sup> The feasible JN pointwise interval also performs well away from

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<sup>1</sup>Recall that, in our simulation design,  $\hat{\Sigma}_f$  uses the correctly specified dynamic model for  $f_t$ .

the origin but can undercover near zero, particularly when the factor is highly persistent and betas vary over time (black dashed line). The uniform-in- $\beta$  procedure eliminates this sharp deterioration (orange solid line). Its minimum coverage across the interior grid ranges from 93.2% to 95.6%, compared with 72.1% to 90.5% for pointwise inference. This contrast illustrates the distinction between pointwise validity at each fixed evaluation point in Corollary 4.6.1 and the uniform-in- $\beta$  coverage guarantee discussed in Remark 4.7 of the main text. The uniform-in- $\beta$  intervals are conservative at the origin, where empirical coverage is 100% in every design. Comparing the left and right columns shows that allowing for state-dependent idiosyncratic volatility has only a modest effect on coverage. Figure SA-IV.1.3 reports the corresponding average confidence interval lengths for all three procedures. The feasible and oracle JN pointwise intervals have nearly identical average lengths throughout the evaluation grid. The uniform-in- $\beta$  adjustment increases interval length in a neighborhood of the origin, where the pointwise procedure can undercover. Outside that neighborhood, the uniform-in- $\beta$  and pointwise intervals coincide exactly. Finally, note that  $\hat{\Sigma}_\varepsilon(\beta; 1)$  was never negative at any evaluation point in the simulations across all eight designs.

Figure SA-IV.1.2 reports pointwise coverage for the two Fama-MacBeth procedures we consider. The procedure based on  $(\mathbf{p}, \mathbf{q}) = (0, 0)$  for both point and variance estimation corresponds to the conventional implementation in empirical work. In contrast, the modified procedure is based on  $(\mathbf{p}, \mathbf{q}) = (1, 1)$ . Both procedures exhibit a sharp deterioration at  $\beta = 0$ . Coverage at the origin ranges from 26.0% to 39.2% for the procedure based on  $(\mathbf{p}, \mathbf{q}) = (0, 0)$  and from 60.7% to 85.7% for the modified procedure based on  $(\mathbf{p}, \mathbf{q}) = (1, 1)$ . Thus, the modified procedure performs substantially better near the origin, but it remains below the nominal level. This undercoverage is consistent with the discussion following Theorem 4.6 of the main text, which explains why the FM variance estimator is not generally valid at the origin. Figure SA-IV.1.4 reports the corresponding average confidence interval lengths. The curves for the two Fama-MacBeth procedures are nearly indistinguishable in every design. This similarity is consistent with Theorems 4.1 and 4.6 of the main text, which show that, at each fixed nonzero evaluation point,  $\mathbf{q} = 0$  and  $\mathbf{q} = 1$  have the same leading asymptotic variance and the same first-order Fama-MacBeth variance target. Consequently, the improvement from the modified procedure near the origin reflects the use of the higher-order point estimator rather than wider confidence intervals.

Finally, Figure SA-IV.1.5 directly compares the average lengths of the JN uniform-in- $\beta$  intervals and the modified Fama-MacBeth pointwise intervals. Away from the origin, the Fama-MacBeth intervals are wider, especially when the factor is highly persistent. Their greater width is reflected in the overcoverage shown in Figure SA-IV.1.2. As shown in Theorem 4.6 of the main text, this difference arises because the FM variance estimator includes period-by-period variation in conditional expected returns that is not part of the estimation uncertainty around the SACER. Near the origin, the Fama-MacBeth intervals can instead be shorter, but Figure SA-IV.1.2 shows that this ostensible gain in precision is accompanied by substantial undercoverage. Taken together, the results support the uniform-in- $\beta$  JN procedure as a reliable approach to inference across both zero and nonzero evaluation points.

Figure SA-IV.1.1: **Coverage probability.** Each panel reports empirical coverage at the 20 interior evaluation points for one of the eight designs in Table SA-IV.1.1. The black dashed line is the feasible JN pointwise interval, the orange solid line is the feasible JN uniform-in- $\beta$  interval using the default calibration of  $\hat{\delta}$ , and the blue solid line is the infeasible oracle JN pointwise interval. The horizontal grey dotted line denotes the nominal coverage probability of 0.95. These results are based on 1,000 Monte Carlo replications per design.

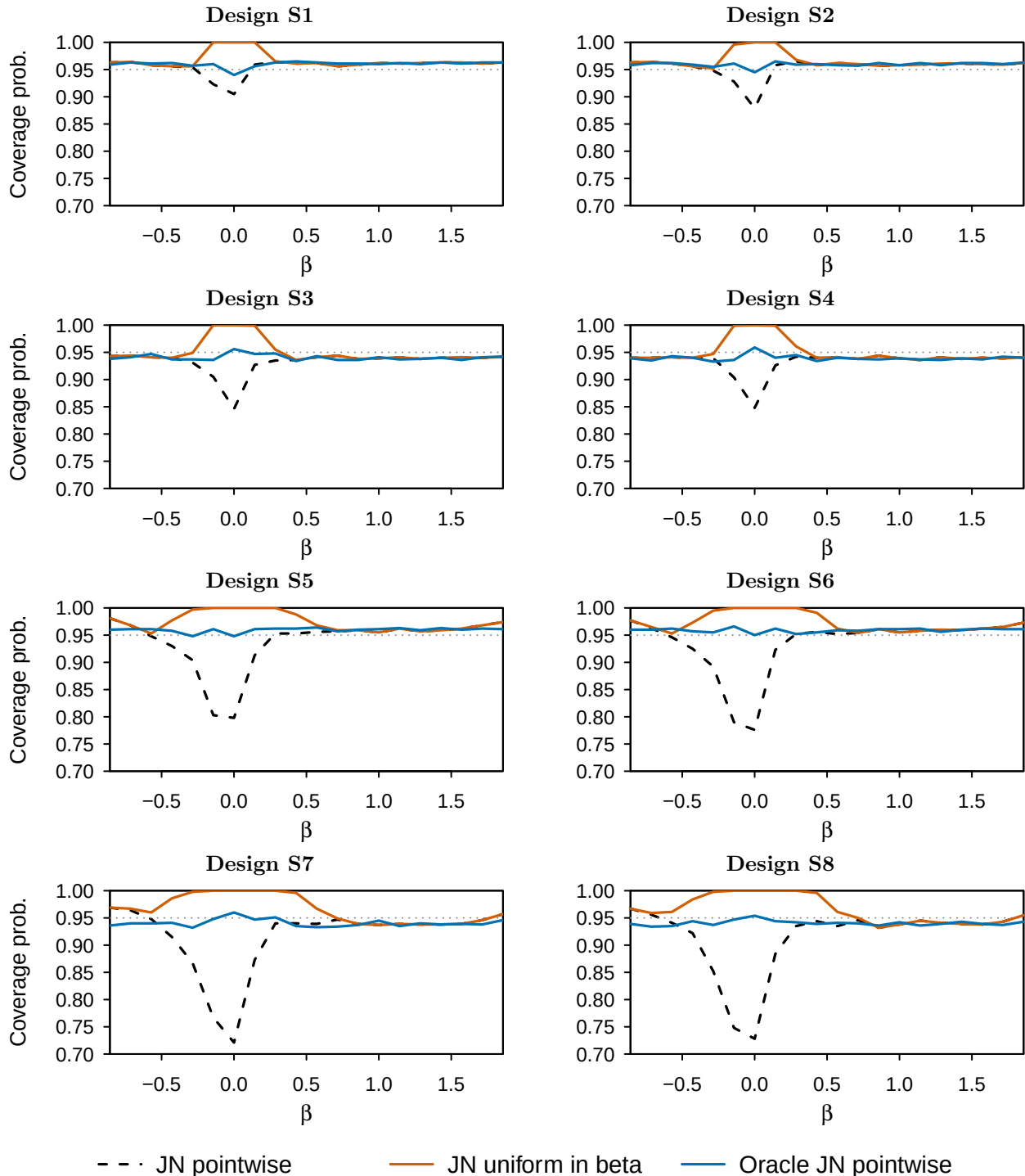


Figure SA-IV.1.2: **Pointwise coverage using Fama-MacBeth variance estimators.** Each panel reports empirical coverage at the 20 interior evaluation points for one of the eight designs in Table SA-IV.1.1. The blue solid line uses the  $(\mathbf{p}, \mathbf{q}) = (0, 0)$  estimates and the corresponding Fama-MacBeth variance estimator. The orange dashed line is the modified Fama-MacBeth procedure based on  $(\mathbf{p}, \mathbf{q}) = (1, 1)$ . The horizontal grey dotted line denotes the nominal coverage probability of 0.95. These results are based on 1,000 Monte Carlo replications per design.

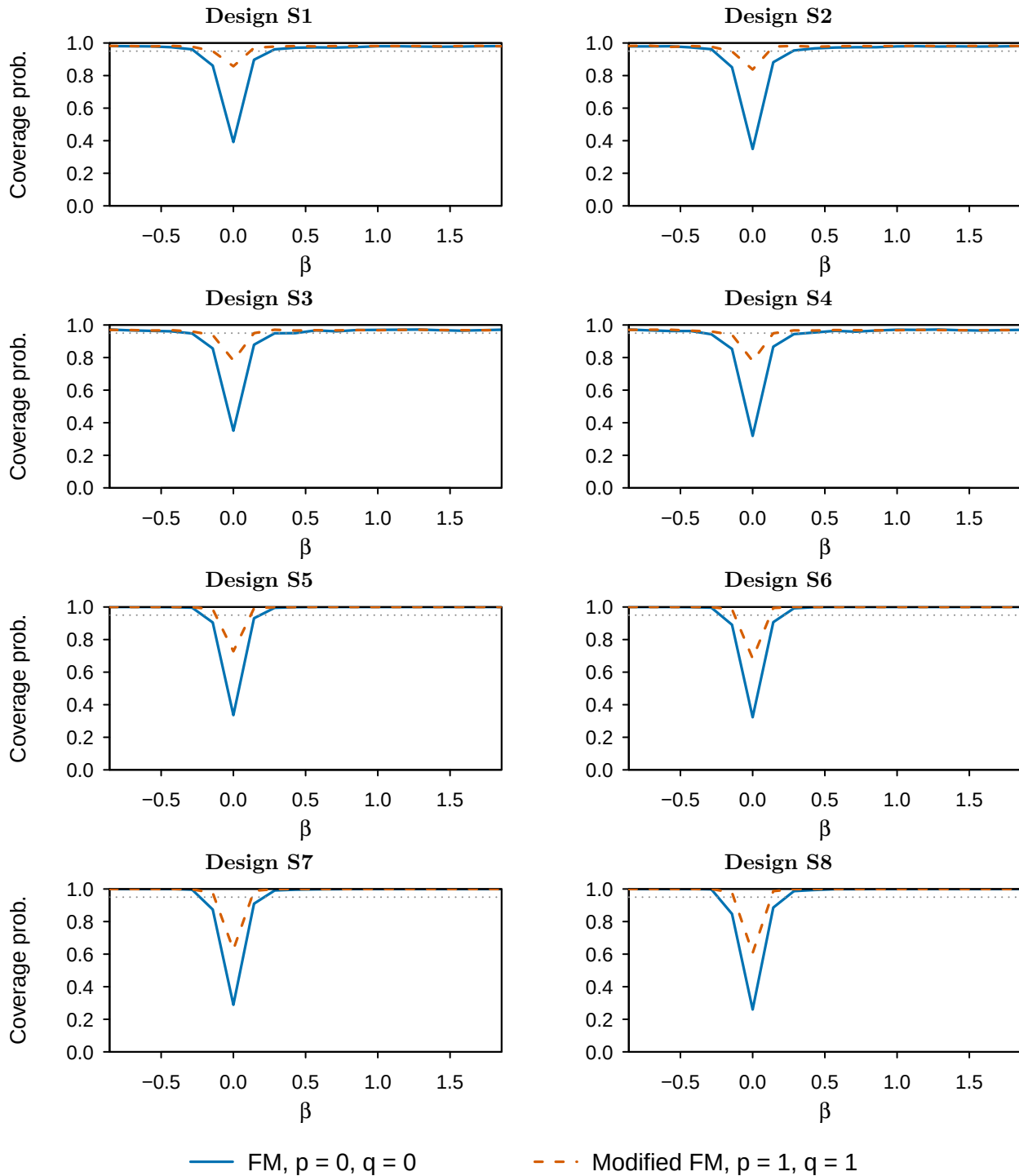


Figure SA-IV.1.3: **Average confidence-interval length.** Each panel reports the average length of the 95% confidence interval at the 20 interior evaluation points for one of the eight designs in Table SA-IV.1.1. The black dashed line is the feasible JN pointwise interval, the orange solid line is the feasible JN uniform-in- $\beta$  interval using the default calibration of  $\hat{\delta}$ , and the blue solid line is the infeasible oracle JN pointwise interval. These results are based on 1,000 Monte Carlo replications per design.

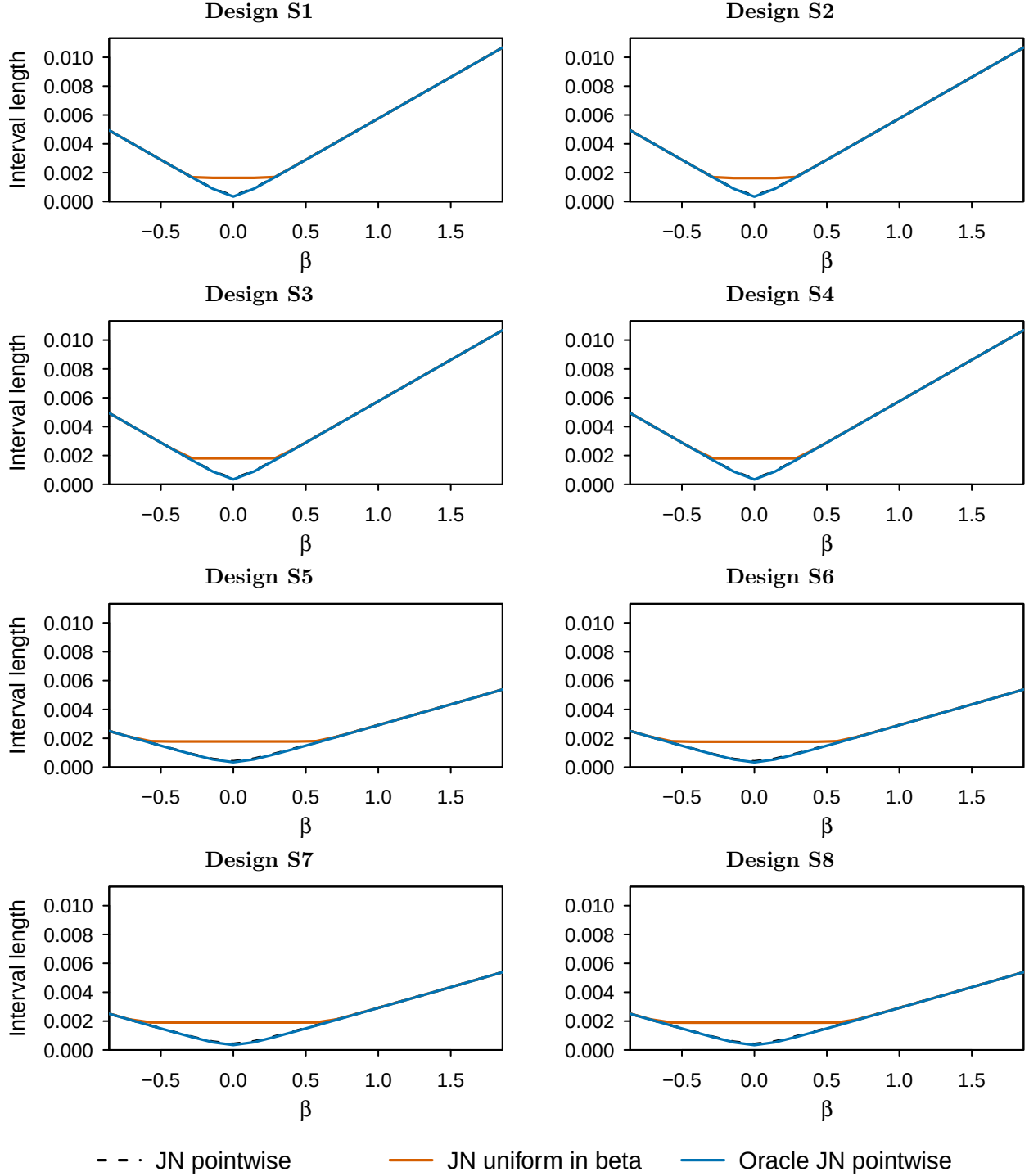


Figure SA-IV.1.4: **Average confidence-interval length using Fama-MacBeth variance estimators.** Each panel reports the average length of the 95% confidence interval at the 20 interior evaluation points for one of the eight designs in Table SA-IV.1.1. The blue solid line uses the  $(p, q) = (0, 0)$  estimates and the corresponding Fama-MacBeth variance estimator. The orange dashed line is the modified Fama-MacBeth procedure based on  $(p, q) = (1, 1)$ . These results are based on 1,000 Monte Carlo replications per design.

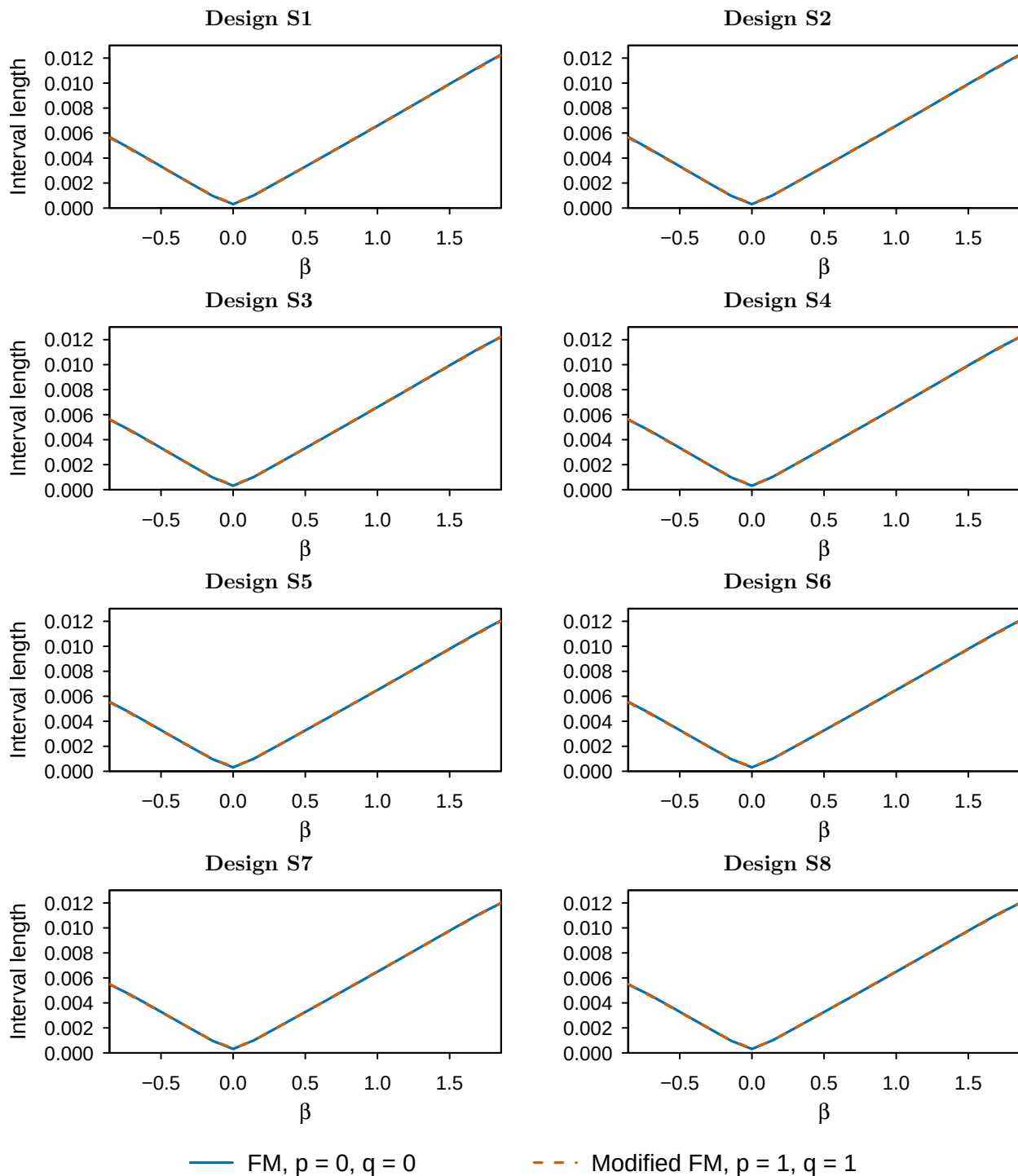
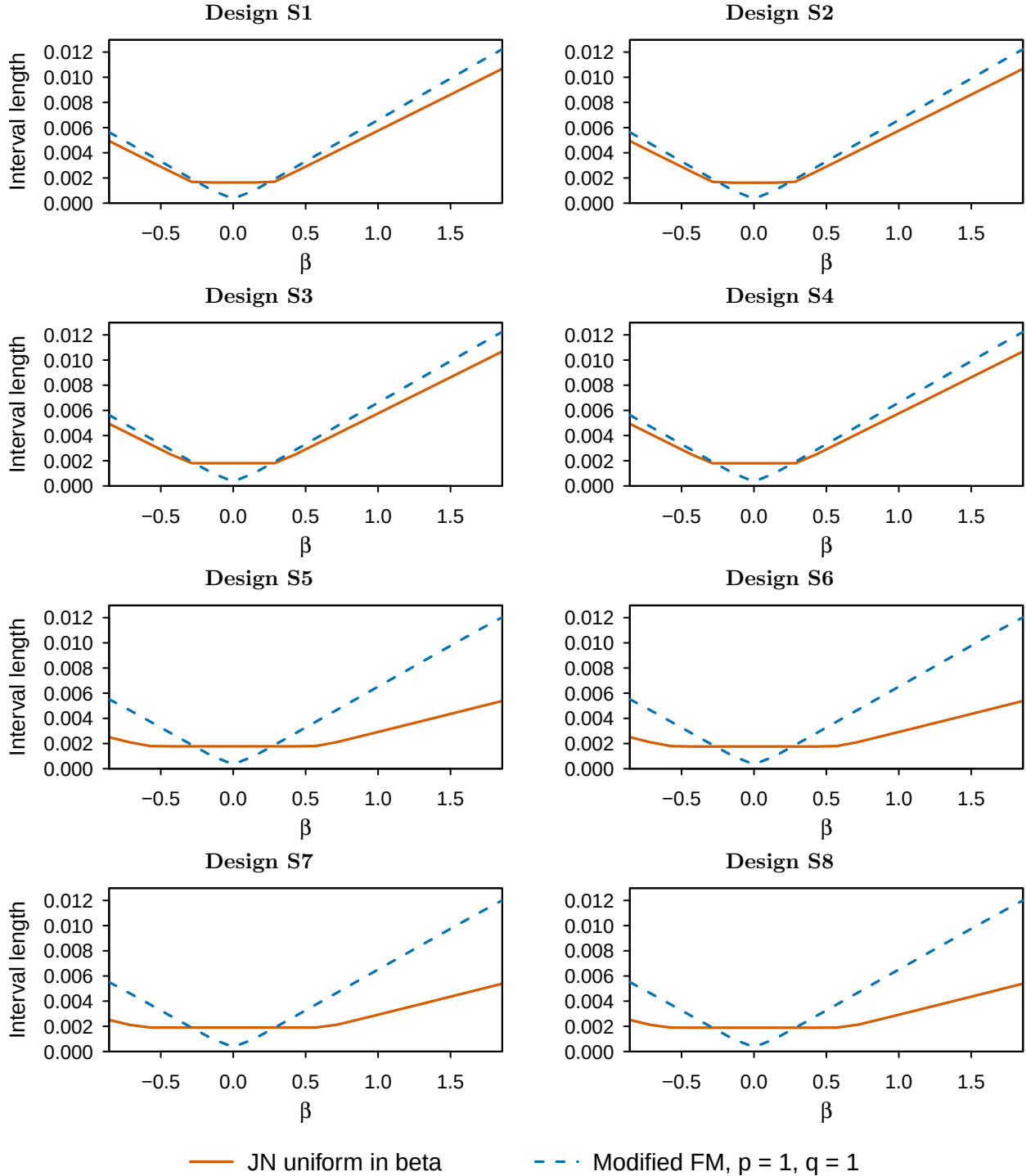


Figure SA-IV.1.5: **Average confidence-interval length: JN uniform-in- $\beta$  and modified Fama-MacBeth.** Each panel reports the average length of the 95% confidence interval at the 20 interior evaluation points for one of the eight designs in Table SA-IV.1.1. The orange solid line is the JN uniform-in- $\beta$  interval using the default calibration of  $\hat{\delta}$ . The blue dashed line is the modified Fama-MacBeth pointwise interval. Both procedures use the same feasible estimates. Results are based on 1,000 Monte Carlo replications per design.



## SA-IV.2 Empirical Applications

In this section, we provide additional results illustrating our new procedures. First, we provide additional evidence and implementation details for the business-credit factor introduced in the main text. Second, we revisit the capital share risk factor of [Lettau et al. \(2019\)](#), providing another application of our new estimation and inference procedures.

### SA-IV.2.1 Additional Results for Business Credit Cycle Factor

#### SA-IV.2.1.1 Out-of-Sample Predictive Regressions

As an additional check on the predictive performance of the business-credit factor, Table [SA-IV.2.1](#) reports out-of-sample  $R^2$  from expanding-window predictive regressions. The out-of-sample  $R^2$  compares the mean squared forecast error from each predictive regression with that from the recursively estimated historical mean forecast. We use data through 1989 as the initial training sample and evaluate the forecasts from January 1990 through December 2019. The business-credit factor produces positive out-of-sample  $R^2$  at all four horizons and outperforms each of the benchmark predictors.

**Table SA-IV.2.1: Out-of-Sample Predictive Regressions** This table reports out-of-sample  $R^2$  (in percent) from expanding window predictive regressions of cumulative market excess returns on an individual predictor from [Welch and Goyal \(2008\)](#) for horizons of 1, 3, 6, and 12 months ahead. The row labeled “CGP” reports results for the SLOOS only portion of our risk factor as studied in [Chava et al. \(2015\)](#). The row labeled “BCF” reports results for our proposed risk factor. Positive values have been bolded. The training period is 1965m1–1989m12 and the evaluation sample is 1990m1–2019m12.

|                             | $\ell = 1$  | $\ell = 3$  | $\ell = 6$   | $\ell = 12$  |
|-----------------------------|-------------|-------------|--------------|--------------|
| (log) Dividend Price Ratio  | -1.79       | -5.51       | -12.00       | -27.91       |
| (log) Dividend Yield        | -1.94       | -5.35       | -12.12       | -26.96       |
| (log) Earnings Price Ratio  | -0.90       | -3.70       | -8.55        | -14.87       |
| (log) Dividend Payout Ratio | -1.50       | -4.51       | -6.51        | -5.08        |
| Stock Variance              | <b>0.79</b> | -4.15       | -1.54        | -0.35        |
| Book-to-Market Ratio        | -0.60       | -2.01       | -4.68        | -11.79       |
| Net Equity Expansion        | -2.35       | -7.20       | -16.86       | -24.49       |
| Treasury Bill Yield         | <b>0.02</b> | -0.08       | -0.38        | -0.55        |
| Long Term Treasury Yield    | -0.03       | -0.77       | -2.51        | -8.44        |
| Long Term Treasury Return   | -0.14       | -1.10       | -0.84        | -0.98        |
| Term Spread                 | -0.98       | -2.12       | -1.68        | <b>4.26</b>  |
| Default Yield Spread        | -2.43       | -5.89       | -6.84        | -5.46        |
| Default Return Spread       | -0.57       | -1.09       | -0.43        | -0.95        |
| (lagged) Inflation          | -0.61       | -0.08       | <b>2.56</b>  | <b>2.58</b>  |
| CGP                         | <b>1.67</b> | <b>3.98</b> | <b>4.41</b>  | <b>3.88</b>  |
| BCF                         | <b>3.00</b> | <b>9.60</b> | <b>13.90</b> | <b>11.88</b> |

#### SA-IV.2.1.2 Implementation and Tuning

Here we provide further details on the implementation of our main empirical application. The values of  $H_{it}$  and  $J_t$  are obtained following the procedure described in Section [SA-III.1](#). The grid used to select  $J_t$  is described in Section [SA-III.4](#). Figure [SA-IV.2.1](#) summarizes the resulting data-driven tuning choices and the

effective cross-sectional sample size over time. The selected first-step windows vary substantially across firms and months:  $H_{it}$  equals its lower bound for 15.5% and its upper bound for 2.2% of firm-month observations assigned to portfolios. The selected number of portfolios ranges from 5 to 40 and reaches its upper bound in 4.8% of the sample months.

The effective cross-sectional sample size ranges from 2,153 to 5,122 firms, with an average of 3,949 firms per month. Even when the upper bound on  $J_t$  is attained, the smallest realized portfolio contains 54 firms. The sharp increase in the effective cross-sectional sample size in January 1976 reflects the beginning of CRSP's Nasdaq coverage in December 1972. Because many Nasdaq firms have their first usable return in January 1973, they first satisfy the requirement of 36 prior monthly observations in January 1976 and then enter the portfolio sorts.

Finally, the uniform-in- $\beta$  calibration described in Section SA-III.4 gives  $\hat{\delta} = 0.0151$ . With this threshold, the uniform variance floor applies at the seven reporting grid points from  $\beta = -0.75$  through  $\beta = 0.75$ .

## SA-IV.2.2 Capital Share Risk Factor

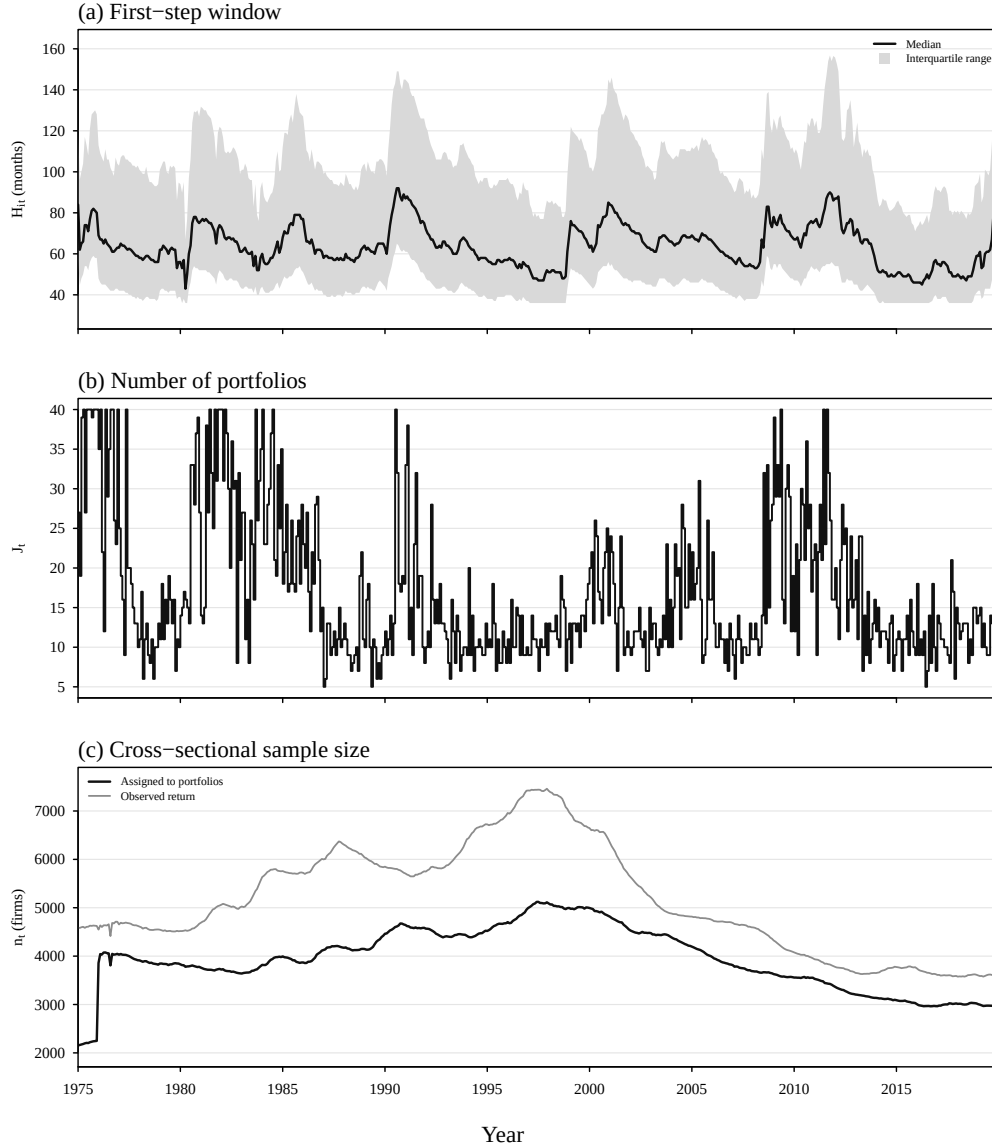
In this section, we revisit the capital share risk factor proposed by Lettau et al. (2019) using our new estimation and inference procedures. Lettau et al. (2019) argue that an empirical model in which growth in the capital share of aggregate income is the single source of macroeconomic risk explains a substantial fraction of the cross-sectional variation in expected returns across a range of equity characteristic portfolios and non-equity asset classes, with positive exposure associated with higher expected returns. Their benchmark capital share measure is constructed from quarterly data on the labor share of the nonfarm business sector from the Bureau of Labor Statistics' Productivity and Costs release.<sup>2</sup> Specifically, they define the capital share as  $KS_t = 1 - LS_t$  and measure its growth over eight quarters. In our implementation, we multiply this log growth rate by 100 and construct the risk factor as  $f_t = 100[\log(KS_t) - \log(KS_{t-8})]$ , where  $LS_t$  denotes the labor share at time  $t$ .<sup>3</sup>

We use the same CRSP stock universe and treatment of delisting returns as in Section 6 of the main text. We form quarterly raw returns by compounding the three monthly returns within each quarter and use the one-factor quarterly implementation choices described in Section SA-III.4. The values of  $H_{it}$  and  $J_t$  are obtained following the procedure described in Section SA-III.1. Applying the default support rule to the point-estimation betas yields a  $J_t$ -selection grid of 25 equally spaced points with endpoints  $-1.92$  and  $1.92$ , while the corresponding rule applied to the inferential betas yields a reporting and inference grid of 11 equally spaced points with endpoints  $-2.37$  and  $2.37$ , where all endpoints are rounded. The full reporting interval lies within the realized cross-sectional range of the  $\hat{\beta}_{it}$  values used for inference in 90.3% of the estimation quarters. Figure SA-IV.2.2 summarizes the resulting data-driven tuning choices and effective cross-sectional sample size over time. The selected first-step window  $H_{it}$  equals its lower bound for 73.3% and its upper bound for 2.5% of firm-quarter observations assigned to portfolios. The selected number of portfolios ranges from 6 to 40 and reaches its upper bound in 1.3% of the sample quarters. The effective cross-sectional sample size ranges from 1,269 to 2,905 firms, with an average of 2,285 firms per quarter; the smallest realized portfolio contains 40 firms. Finally, the uniform-in- $\beta$  calibration gives  $\hat{\delta} = 0.0548$ . The

<sup>2</sup>We obtain these data from the paper's replication package available at <https://onlinelibrary.wiley.com/doi/full/10.1111/jofi.12772>.

<sup>3</sup>Gospodinov and Robotti (2021) reexamine these findings and argue that the original evidence of statistical significance may reflect the persistence induced by using overlapping long-horizon returns and capital-share growth. These authors recommend using nonoverlapping data or alternative empirical methods. This critique motivates our alternative implementation: we retain the eight-quarter capital-share growth factor of Lettau et al. (2019) but use nonoverlapping one-quarter returns and our beta-sorted portfolio procedure rather than their two-pass risk-premium procedure based on overlapping returns.

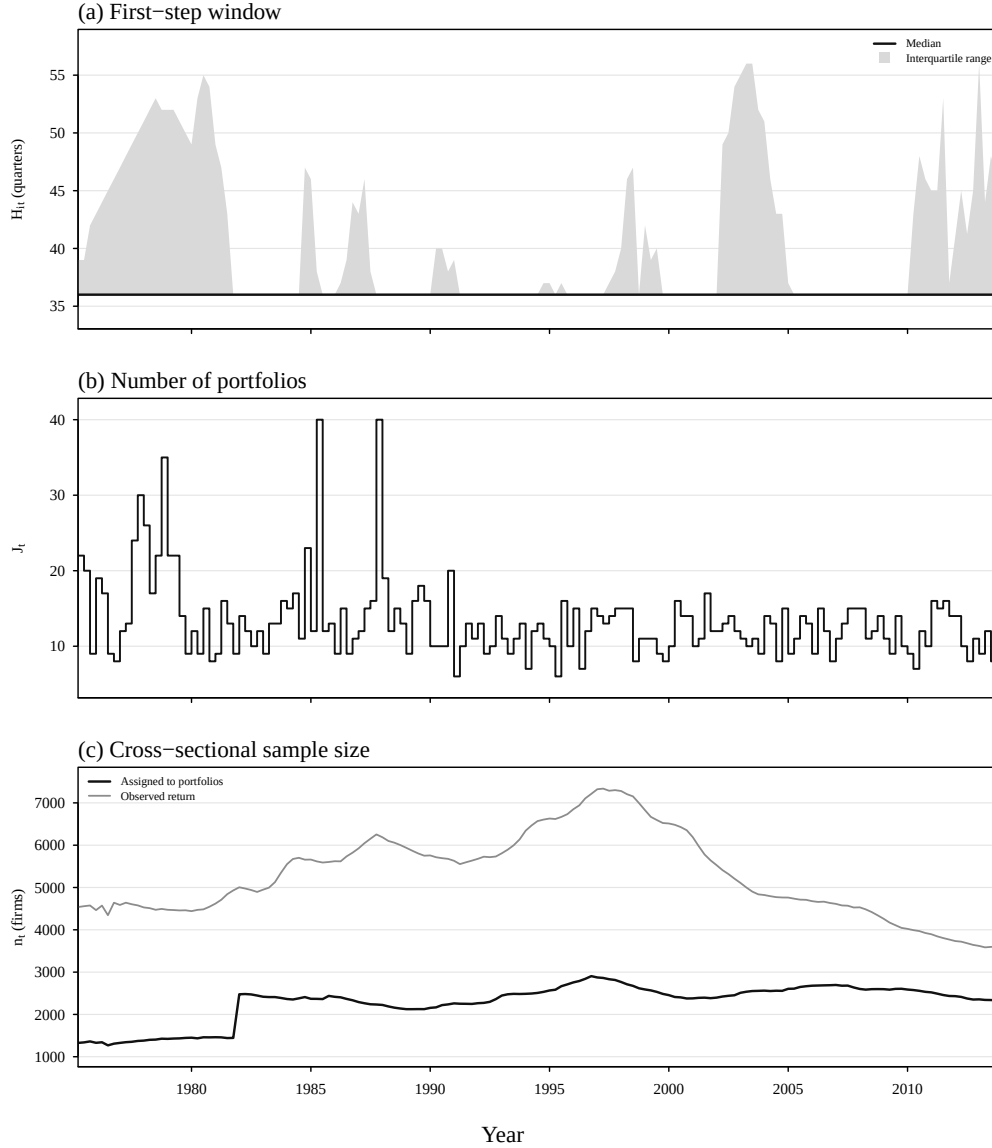
Figure SA-IV.2.1: **Tuning parameters and effective sample size over time.** Panel A reports the monthly median of the selected first-step window  $H_{it}$  across firms, with the shaded region denoting the interquartile range. Panel B reports the selected number of portfolios,  $J_t$ . Panel C reports the effective cross-sectional sample size,  $n_t$ , defined as the number of firms with a finite beta estimate that are assigned to a portfolio; the grey line reports the number of firms with an observed return before imposing the first-step estimation requirement. The sample period is January 1975 through December 2019.



estimated risk-factor variance component reaches this threshold at approximately  $\beta = -1.27$  and  $\beta = 1.27$ . Accordingly, the uniform variance floor applies at the five reporting grid points from  $\beta = -0.95$  through  $\beta = 0.95$ .

Figure SA-IV.2.3 reports the estimated SACER and associated inference using the capital share risk factor. The point estimate based on  $(p, q) = (0, 0)$  is given by the black line. The estimated relation generally slopes upward; specifically, the estimated SACER increases from 4.17% per quarter at  $\beta = -2.37$  to 5.02% at  $\beta = 2.37$ . This pattern is qualitatively consistent with the positive price of capital share risk

Figure SA-IV.2.2: **Tuning parameters and effective sample size over time for the capital share application.** Panel A reports the quarterly median of the selected first-step window  $H_{it}$  across firms, with the shaded region denoting the interquartile range. Panel B reports the selected number of portfolios,  $J_t$ . Panel C reports the effective cross-sectional sample size,  $n_t$ , defined as the number of firms with a finite beta estimate that are assigned to a portfolio; the grey line reports the number of firms with an observed complete quarterly return before imposing the first-step estimation requirement. The estimation period is 1975Q2 through 2013Q4.



documented by Lettau et al. (2019). The corresponding high-minus-low contrast point estimate based on  $(\mathbf{p}, \mathbf{q}) = (0, 0)$  is 0.85 percentage points. Table SA-IV.2.2 reports inference for the same contrast using the  $(\mathbf{p}, \mathbf{q}) = (1, 1)$  estimate; consequently, the confidence intervals need not be centered on the black-line estimate. The JN standard error is 1.030, giving a two-sided  $p$ -value of 0.605, while the modified FM standard error is 0.634, giving a two-sided  $p$ -value of 0.401. The  $\hat{\Sigma}_f$ -only benchmark also fails to reject, so none of the three procedures rejects a zero contrast.

Figure SA-IV.2.3: **Inference on the SACER.** The black line reports the  $(\mathbf{p}, \mathbf{q}) = (0, 0)$  estimate of the SACER. The grey region is the 95% JN pointwise interval. The red dashed lines are the 95% JN uniform-in-beta intervals, and the green dotted lines are the 95% modified FM pointwise intervals. All three intervals use  $(\mathbf{p}, \mathbf{q}) = (1, 1)$ . Results are reported for quarterly raw returns (in percent), formed by compounding the three monthly returns within each quarter. The sample period is 1965Q2–2013Q4 with a 40-quarter burn-in period, so the estimation period is 1975Q2–2013Q4.

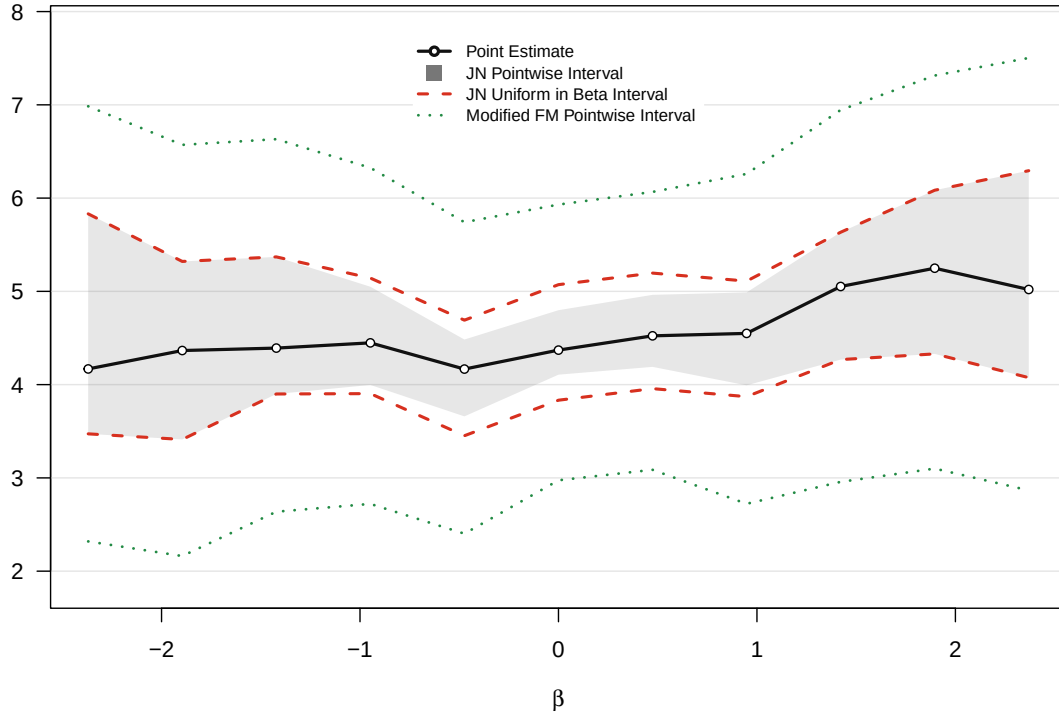


Table SA-IV.2.2: **High Minus Low Contrast.** The table reports inference for  $\hat{\mu}(2.37) - \hat{\mu}(-2.37)$  in quarterly percentage points. Each row reports the standard error, 90% confidence interval, and two-sided normal-approximation  $p$ -value under the JN, modified FM, or  $\hat{\Sigma}_f$ -only variance estimator. The confidence intervals are centered at the  $(\mathbf{p}, \mathbf{q}) = (1, 1)$  estimate.

| Variance estimator    | SE    | 90% CI            | $p$ -value |
|-----------------------|-------|-------------------|------------|
| JN                    | 1.030 | $[-1.162, 2.227]$ | 0.605      |
| Modified FM           | 0.634 | $[-0.511, 1.576]$ | 0.401      |
| $\hat{\Sigma}_f$ only | 0.874 | $[-0.904, 1.969]$ | 0.542      |

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